

Sri Lanka Institute of Information Technology



Software Engineering | SE 2030

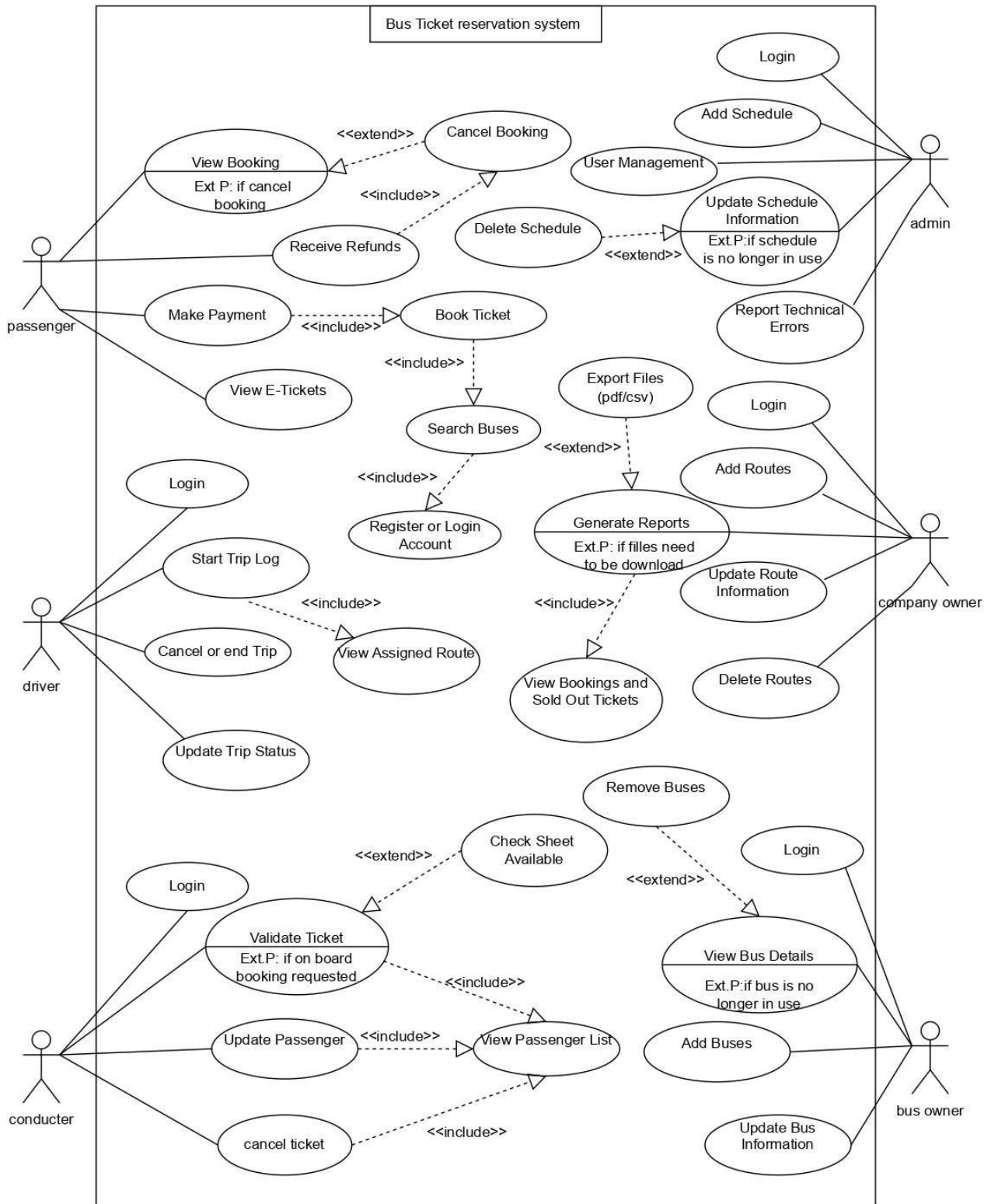
Lab Sheet 05

2025-Y2-S1-KU-27

Topic Name: Web-based Bus Ticket Reservation System

Reg No	Name
IT24104374	Zikra.G.F
IT24102866	Kalendra G.K.A
IT24103309	Daskon W.W.D.A.M.P.B
IT24103015	Jayawardhane T.D
IT24103931	Dissanayake Y.I
IT24102501	Bandara C.V

User Case Diagram



IT24103015- Jayawardhane T D
Use Case Scenarios for Actor: Passenger

Use Case PAS-01: Login to System

Number	PAS-01
Name	Login to System
Summary	A passenger logs into the system to access booking features.
Priority	5
Preconditions	The passenger has a registered account.
Postconditions	The passenger is authenticated and redirected to their dashboard.
Primary Actor(s)	Passenger
Secondary Actor	-

Trigger	User navigates to the login page.	
Main Scenario	Step	Action
	1	System displays the login form.
	2	Passenger enters their email/username and password.
	3	Passenger clicks the 'Login' button.
	4	System validates the credentials.
	5	System authenticates the user and redirects them to the passenger dashboard.
Extensions	Step	Branching Action
	4a	Condition: Invalid credentials. Action: System displays an "Invalid login credentials" error.
	4b	Condition: User clicks "Forgot Password". Action: System initiates the password reset process (a separate minor use case).

Use Case PAS-02: Search for Available Buses

Number	PAS-02
Name	Search for Available Buses
Summary	A passenger searches for buses based on route, date, and time.
Priority	5
Preconditions	Passenger is logged in.
Postconditions	A list of available buses matching the criteria is displayed.
Primary Actor(s)	Passenger
Secondary Actor	-

Trigger	Passenger navigates to the search page.	
Main Scenario	Step	Action
	1	Passenger enters search criteria (departure, destination, date, time).
	2	Passenger clicks the 'Search' button.
	3	System validates the input and queries the schedule and seat availability databases.
	4	System displays a list of available buses with details (time, price, seat count)
Extensions	Step	Branching Action
	3a	Condition: Invalid or incomplete search parameters. Action: System prompts the user to correct the input.
	4a	Condition: No buses are found. Action: System displays a "No buses available" message.

Use Case PAS-03: Book a Ticket

Number	PAS-03
Name	Book a Ticket
Summary	A passenger reserves and pays for a seat on a selected bus.
Priority	5
Preconditions	1. Passenger is logged in. 2. Passenger has searched and selected an available bus.
Postconditions	1. A new ticket is created. 2. Seat availability is updated. 3. Passenger receives a confirmation.
Primary Actor(s)	Passenger
Secondary Actor	Payment Gateway (Simulated)

Trigger	Passenger clicks 'Book Now' for a selected bus.	
Main Scenario	Step	Action
	1	System displays the seat map for the selected bus.
	2	Passenger selects an available seat.
	3	Passenger proceeds to checkout and provides payment details.
	4	System simulates payment processing.
	5	System generates a ticket, updates all databases, and sends a confirmation email.
	6	System displays the e-ticket to the passenger
Extensions	Step	Branching Action
	2a	Condition: Selected seat is taken. Action: System displays an error and refreshes the seat map.
	4a	Condition: Simulated payment fails. Action: System notifies the passenger and returns to the payment step.

Use Case PAS-04: View Booking History

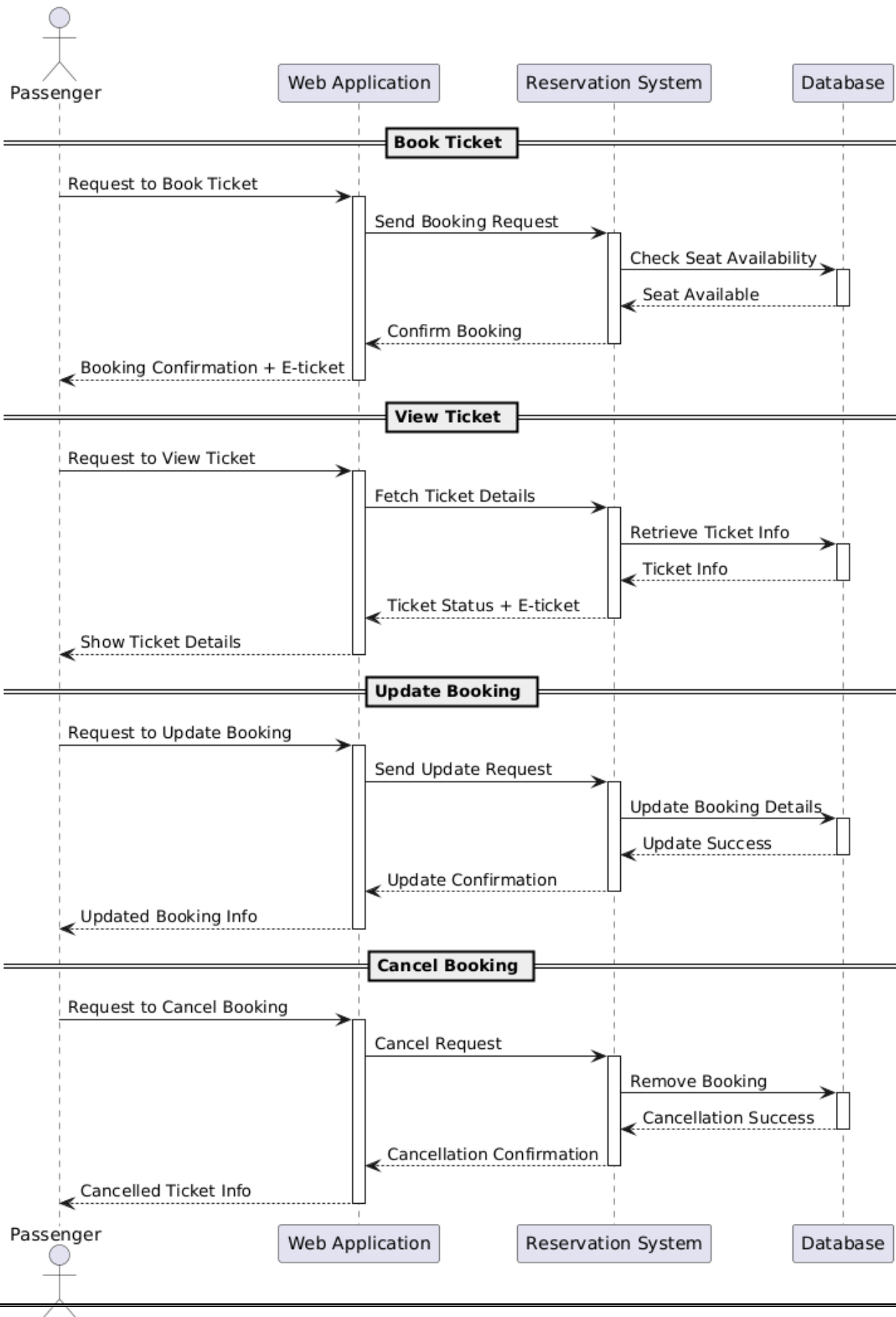
Number	PAS-04
Name	View Booking History
Summary	A passenger views a list of their current and past bookings.
Priority	3
Preconditions	Passenger is logged in.
Postconditions	The passenger's booking history is displayed.
Primary Actor(s)	Passenger
Secondary Actor	-

Trigger	Passenger clicks on 'My Bookings' or 'History' in their dashboard.	
Main Scenario	Step	Action
	1	Passenger navigates to the 'Booking History' section
	2	System retrieves all tickets associated with the passenger's account from the database.
	3	System displays the list of bookings, sorted by date.
Extensions	Step	Branching Action
	3a	Condition: No booking history exists. Action: System displays a "No bookings found" message.

Use Case PAS-05: Cancel a Booking

Number	Pas-05
Name	Cancel a Booking
Summary	A passenger cancels an existing upcoming booking
Priority	4
Preconditions	1. Passenger is logged in. 2. A valid, upcoming booking exists.
Postconditions	1. The booking is cancelled. 2. The seat is released. 3. A refund is processed (simulated).
Primary Actor(s)	Passenger
Secondary Actor	-

Trigger	Passenger selects a booking from their history and chooses to cancel it.	
Main Scenario	Step	Action
	1	Passenger views their booking history and selects an upcoming booking.
	2	Passenger clicks the 'Cancel Booking' button.
	3	System asks for confirmation to cancel.
	4	Passenger confirms the cancellation.
	5	System cancels the ticket, updates the seat status, and simulates a refund.
	6	System sends a cancellation confirmation email to the passenger.
Extensions	Step	Branching Action
	5a	Condition: The departure time is too close (per system policy). Action: System displays a "Cancellation not allowed at this time" error.



IT24104374-Zikra G.F

Use Case Scenarios for Actor: Bus Owner

Case BO-01: Login to System

Number	BO-01
Name	Login to System
Summary	The bus owner provides credentials to gain authenticated access to the system dashboard.
Priority	5
Preconditions	1. The bus owner has a valid user account in the system. 2. The bus owner is not already logged in.
Postconditions	1. The bus owner is authenticated. 2. The system dashboard for the conductor role is displayed.
Primary Actor(s)	Bus owner
Secondary Actor	-

Trigger	The bus owner navigates to the system login page.	
Main Scenario	Step	Action
	1	System displays the login form with fields for username and password.
	2	Bus owner enters their assigned username and password.
	3	Bus owner clicks the 'Login' button.
	4	System validates the credentials against the user database.
	5	System confirms credentials are valid and grants access
	6	System redirects the bus owner to their role-specific dashboard.
Extensions	Step	Branching Action
	4a	Condition: Username is not found. Action: System displays an "Invalid username or password" error.
	4b	Condition: Password is incorrect. Action: System displays an "Invalid username or password" error.

Use Case BO-02: Add new bus

Number	BO-02
Name	Add new Bus
Summary	The bus owner adds new buses to the system to expand their fleet.
Priority	5
Preconditions	1. The bus owner is logged in. 2. The bus owner has access rights to manage buses.
Postconditions	1. The new bus is registered in the system. 2. The fleet is updated with the new bus.
Primary Actor(s)	Bus Owner
Secondary Actor	-

Trigger	The bus owner selects 'Add New Bus'.	
Main Scenario	Step	Action
	1	Owner navigates to the bus management dashboard.
	2	Owner selects 'Add New Bus'.
	3	System displays the add bus form.
	4	Owner enters bus details (e.g., license info, seating capacity).
	5	Owner submits the form.
	6	System validates inputs and registers the new bus.
	7	System confirms the bus has been added.
Extensions	Step	Branching Action
	4a	Condition: Invalid or missing details. Action: System displays an error and requests correction.

Use Case BO-03: View bus Details

Number	BO-03
Name	View Bus Details
Summary	The bus owner views all buses to monitor their operational fleet.
Priority	5
Preconditions	1. The bus owner is logged in.
Postconditions	1. . The system displays details of all buses.
Primary Actor(s)	Bus Owner
Secondary Actor	-

Trigger	The bus owner selects 'View Bus Details'.	
Main Scenario	Step	Action
	1	Owner navigates to the dashboard.
	2	Owner selects 'View Bus Details'.
	3	System retrieves all bus records.
	4	System displays bus information (capacity, license info, status).
	5	Owner reviews the fleet details.
Extensions	Step	Branching Action
	3a	Condition: No buses are available. Action: System displays an empty list message.

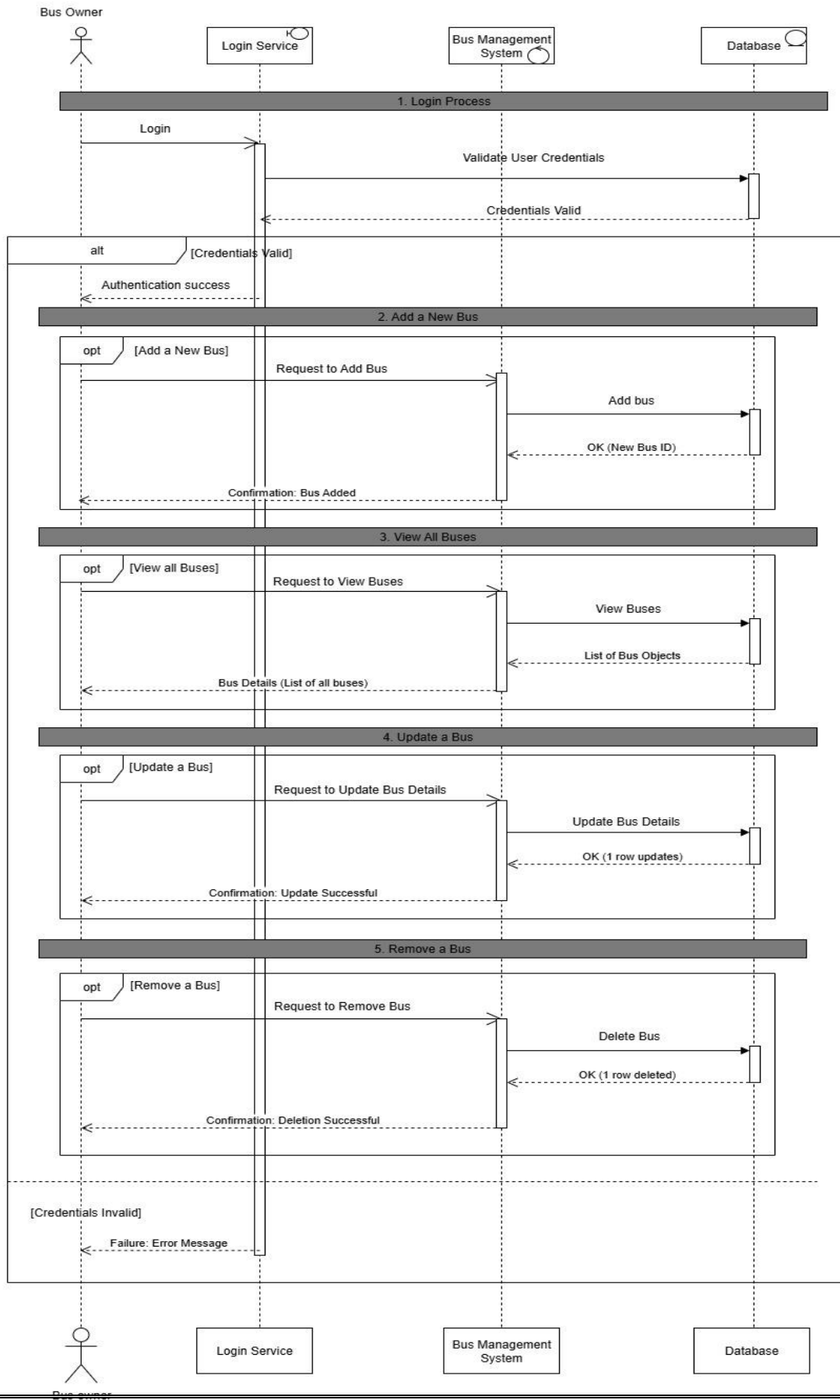
BO-04: Update Bus Details

Number	BO-04	
Name	Update Bus Details	
Summary	The bus owner updates bus details such as seating capacity or license information.	
Priority	4	
Preconditions	1. The bus owner is logged in. 2. The bus exists in the system.	
Postconditions	1. The bus record is updated with new detail	
Primary Actor(s)	Bus Owner	
Secondary Actor	-	
Trigger	The bus owner selects 'Update Bus Details'.	
Main Scenario	Step	Action
	1	Owner navigates to the bus management dashboard.
	2	Owner selects the bus to update.
	3	System displays current details of the bus.
	4	Owner edits seating capacity or license information.
	5	Owner submits the updates.
	6	System validates and saves the changes.
	7	System confirms the bus has been updated.
Extensions	Step	Branching Action
	4a	Condition: Invalid or missing details. Action: System displays an error and requests correction.

BO-05: Remove Bus

Number	BO-05
Name	Remove Bus
Summary	The bus owner removes a bus from the system so that only active buses remain listed.
Priority	4
Preconditions	1. The bus owner is logged in. 2. The bus exists in the system.
Postconditions	1. The bus record is removed from the system. 2. The fleet list contains only active buses.
Primary Actor(s)	Bus Owner
Secondary Actor	-

Trigger	The bus owner selects 'Remove Bus'.	
Main Scenario	Step	Action
	1	Owner navigates to the bus management dashboard.
	2	Owner selects a bus to remove.
	3	System displays the bus details and a remove option.
	4	Owner confirms removal.
	5	System deletes the bus record.
	6	System confirms the bus has been removed.
Extensions	Step	Branching Action
	2a	The selected bus is currently active in a journey. Action: System prevents deletion and shows a warning.



IT24103931-Dissanayake Y.I

Use Case Scenarios for Actor: Bus Conductor

Use Case CON-01: Login to System

Number	CON-01
Name	Login to System
Summary	The bus conductor provides credentials to gain authenticated access to the system dashboard.
Priority	5
Preconditions	1. The conductor has a valid user account in the system. 2. The conductor is not already logged in.
Postconditions	1. The conductor is authenticated. 2. The system dashboard for the conductor role is displayed.
Primary Actor(s)	Bus Conductor
Secondary Actor	-

Trigger	The conductor navigates to the system login page.	
Main Scenario	Step	Action
	1	System displays the login form with fields for username and password.
	2	Conductor enters their assigned username and password.
	3	Conductor clicks the 'Login' button.
	4	System validates the credentials against the user database.
	5	System confirms credentials are valid and grants access
	6	System redirects the conductor to their role-specific dashboard.
Extensions	Step	Branching Action
	4a	Condition: Username is not found. Action: System displays an "Invalid username or password" error.
	4b	Condition: Password is incorrect. Action: System displays an "Invalid username or password" error.

Use Case CON-02: Validate Ticket

Number	CON-02
Name	Validate Ticket
Summary	The conductor checks if a passenger's ticket is valid for the current journey.
Priority	5
Preconditions	1. Conductor is logged in. 2. The bus journey is in progress.
Postconditions	The ticket's validation status is recorded in the log.
Primary Actor(s)	Bus Conductor
Secondary Actor	Passanger

Trigger	Passenger presents a ticket (digital or physical) to the conductor.	
Main Scenario	Step	Action
	1	Conductor selects the 'Validate Ticket' function from their dashboard.
	2	Conductor manually enters the ticket ID.
	3	System checks the ticket database for validity (correct journey, date, time, not cancelled).
	4	System confirms the ticket is valid and displays passenger and seat details.
	5	Conductor informs the passenger that their ticket is valid.
Extensions	Step	Branching Action
	3a	Condition: Ticket is not found or is invalid. Action: System displays an "Invalid Ticket" error. Conductor can then choose to create an on-board ticket (UC CON-03)

Use Case CON-03: Create On-Board Ticket

Number	CON-03
Name	Create On-Board Ticket
Summary	The conductor creates a new ticket for a passenger on the bus if seats are available.
Priority	5
Preconditions	1. Conductor is logged in. 2. The bus journey has started.
Postconditions	1. A new ticket is created. 2. The seat is marked as occupied. 3. Passenger is added to the journey list.
Primary Actor(s)	Bus Conductor
Secondary Actor	Passenger

Trigger	A passenger without a pre-booked ticket requests to board.	
Main Scenario	Step	Action
	1	Conductor selects 'Create On-Board Ticket'.
	2	System displays real-time seat availability for the current journey.
	3	Conductor selects an available seat.
	4	Conductor enters passenger details (e.g., name, contact).
	5	Conductor confirms the ticket creation.
	6	System creates the ticket, updates seat status, and adds passenger to the list.
	7	System provides a ticket confirmation (e.g., printable receipt).
Extensions	Step	Branching Action
	2a	Condition: No seats are available. Action: System displays a "No Seats Available" message.
	4a	Condition: Invalid passenger details entered. Action: System displays an error and prompts for correction.

Use Case CON-04: Update Passenger Status

Number	CON-4
Name	Update Passenger Status
Summary	The conductor marks when a passenger boards or alights the bus.
Priority	4
Preconditions	1. Conductor is logged in. 2. A valid ticket exists for the passenger.
Postconditions	The passenger's status (Boarded/Alighted) is updated in the system.
Primary Actor(s)	Bus Conductor
Secondary Actor	Passenger

Trigger	Passenger physically boards or alights the bus.	
Main Scenario	Step	Action
	1	Conductor views the passenger list for the current journey.
	2	Conductor finds the passenger by name or ticket ID.
	3	Conductor selects the action "Mark as Boarded" or "Mark as Alighted".
	4	System updates the passenger's status and records the time.
		System confirms the update was successful.
Extensions	Step	Branching Action
	2a	Condition: Passenger is not found on the list. Action: . Conductor may need to create a ticket first (UC CON-03).

Use Case CON-05: View Passenger List

Number	CON-5
Name	View Passenger List
Summary	The conductor views the list of all passengers booked on the current journey.
Priority	3
Preconditions	1. Conductor is logged in. 2. A journey is selected or in progress.
Postconditions	The passenger list is displayed.
Primary Actor(s)	Bus Conductor
Secondary Actor	-

Trigger	Conductor navigates to the 'Passenger List' view.	
Main Scenario	Step	Action
	1	Conductor selects the current journey from their active trip list.
	2	Conductor clicks 'View Passenger List'.
	3	System retrieves all passengers with tickets for this journey.
	4	System displays the list, including names, seat numbers, and boarding status.
	5	Conductor can sort or filter the list.
Extensions	Step	Branching Action
	3a	Condition: No passengers are found for the journey. Action: System displays an empty list message.

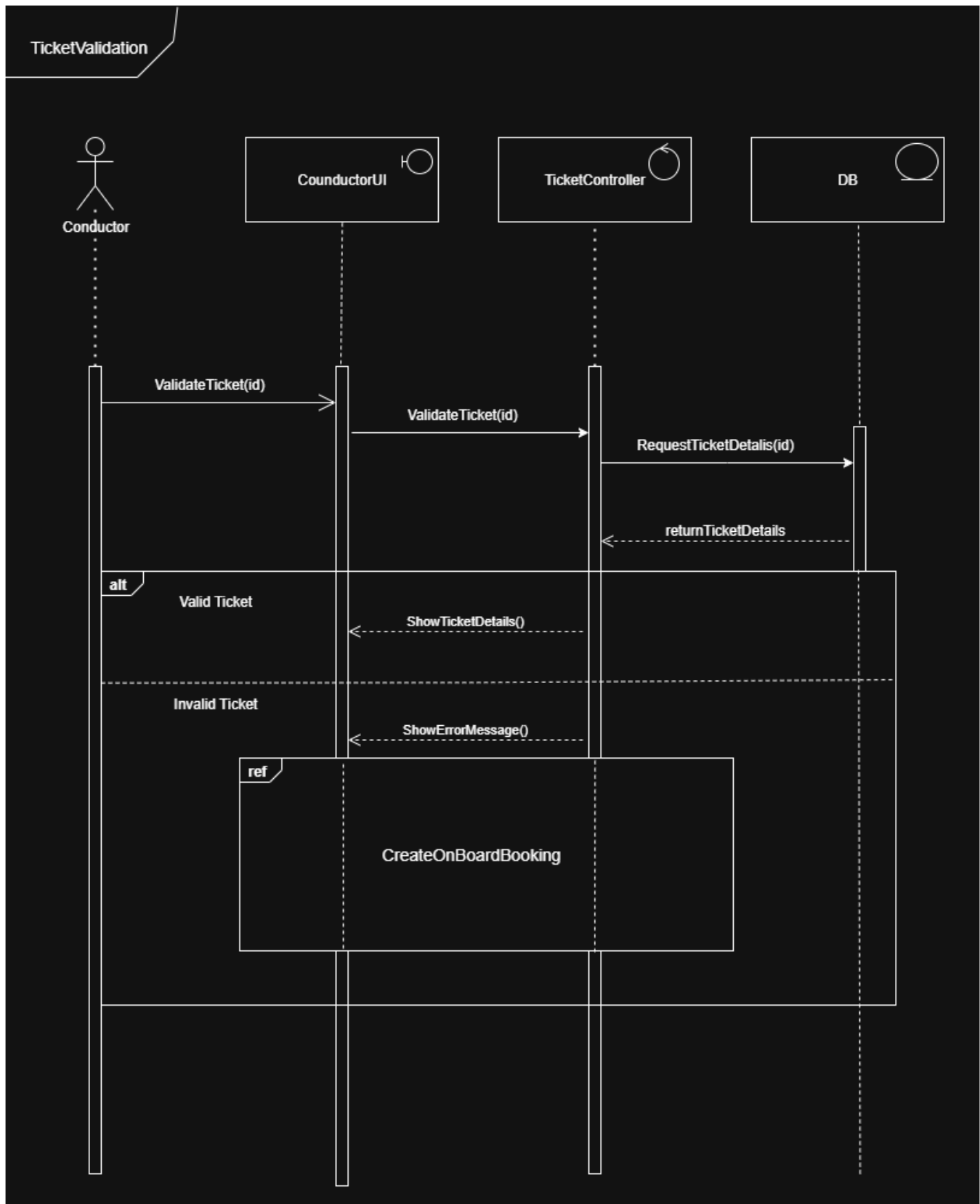
Use Case CON-06: Cancel a Ticket

Number	CON-6
Name	Cancel a Ticket
Summary	The conductor cancels an existing ticket and frees up the seat.
Priority	3
Preconditions	1. Conductor is logged in. 2. A valid ticket exists.
Postconditions	1. The ticket is cancelled. 2. The seat is released and marked as available. 3. Passenger is removed from the journey list.
Primary Actor(s)	Bus Conductor
Secondary Actor	Passenger

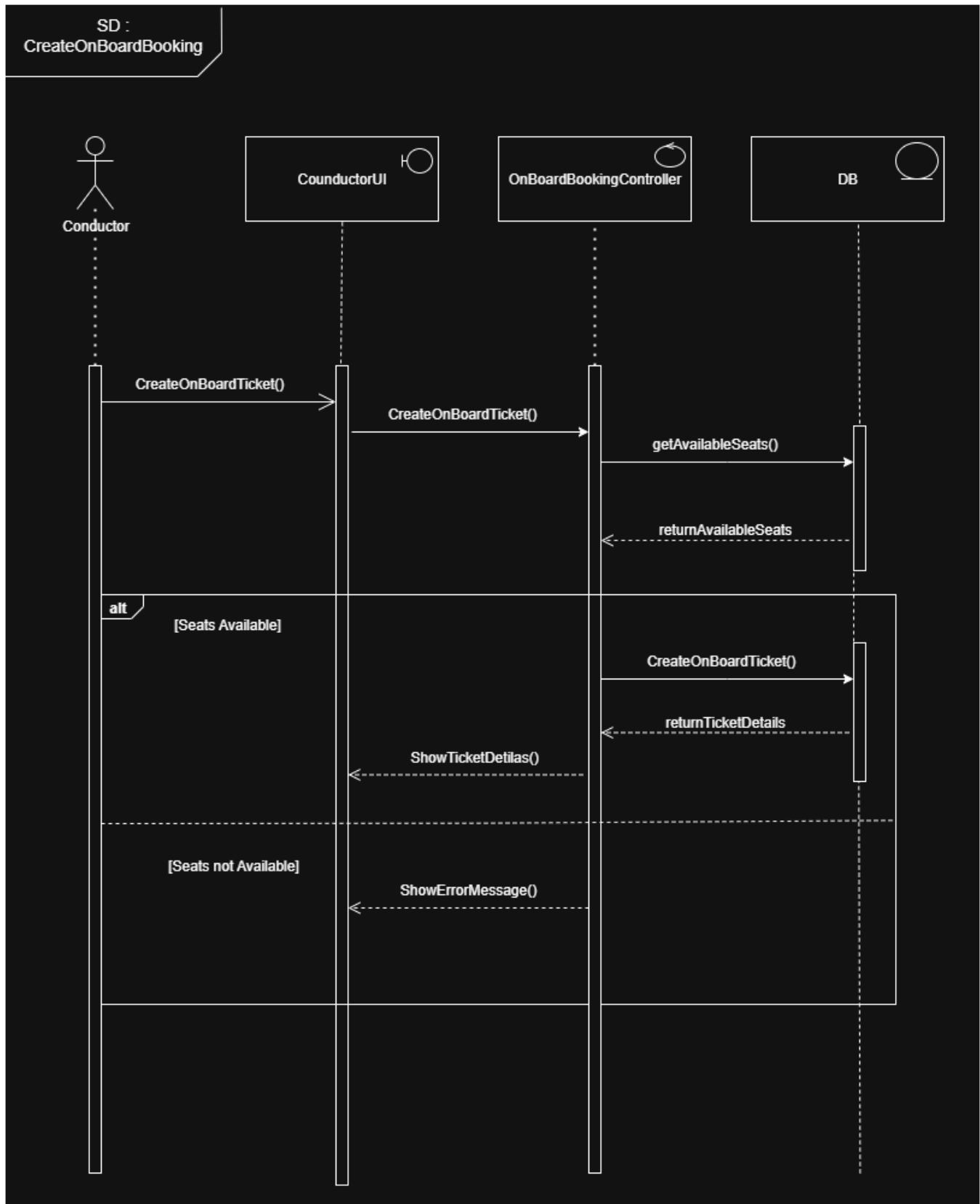
Trigger	A passenger requests to cancel their ticket or is unable to travel.	
Main Scenario	Step	Action
	1	Conductor finds the passenger's ticket via the passenger list or by Searching
	2	Conductor selects the 'Cancel Ticket' option.
	3	System asks for confirmation to cancel.
	4	Conductor confirms the cancellation
	5	System cancels the ticket, updates the seat status to available, and removes the passenger from the list.
	6	System cancels the ticket, updates the seat status to available, and removes the passenger from the list.
Extensions	Step	Branching Action
	1a	Condition: Ticket cannot be found. Action: System displays an error.

Conductor - Sequence Diagram –

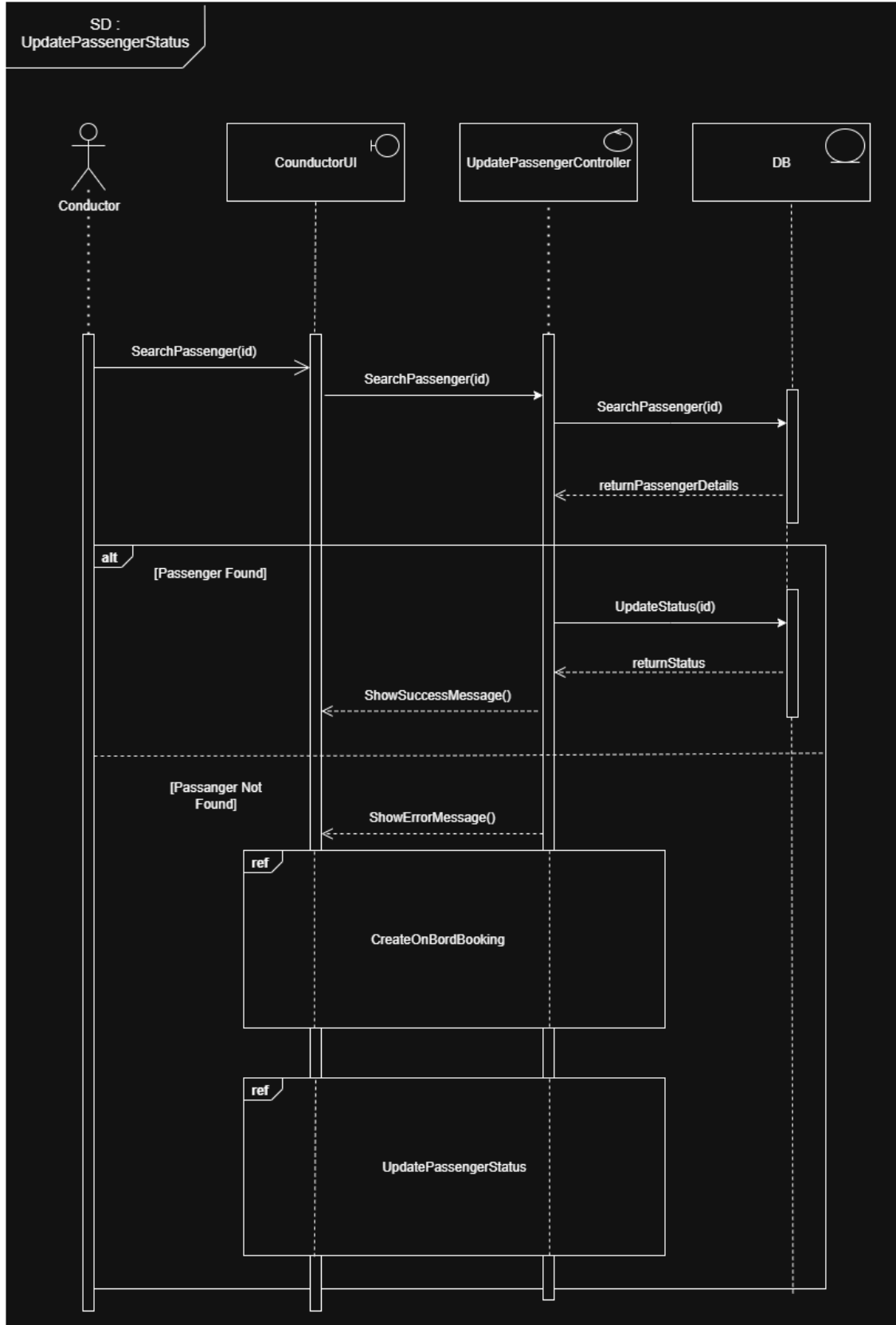
1. Ticket Validations



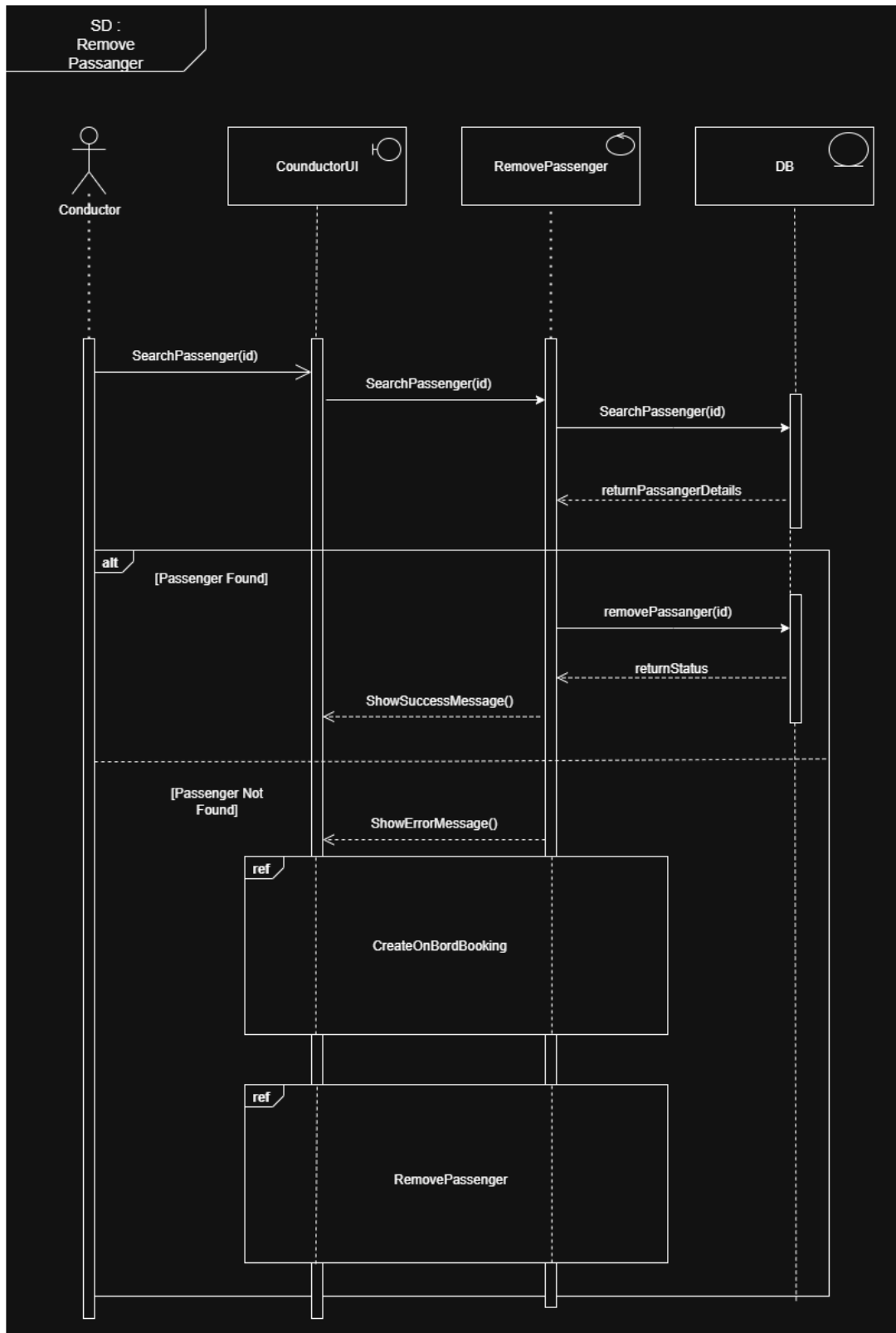
2. On Board Ticket Booking



3. Update Passenger Status (Boarded / alight)



4. Remove Passenger



IT24104374- Daskon W.W.D.A.M.P.B

Use Case Scenarios for Actor: Bus Driver Use

Case DRI-01:Login to System

Number	DRI-01
Name	Login to System
Summary	The bus Driver provides credentials to gain authenticated access to the system dashboard.
Priority	5
Preconditions	1. The Driver has a valid user account in the system. 2. The Driver is not already logged in.
Postconditions	1. The Driver is authenticated. 2. The system dashboard for the Driver role is displayed.
Primary Actor(s)	Bus Driver
Secondary Actor	-

Trigger	The Driver navigates to the system login page.	
Main Scenario	Step	Action
	1	System displays the login form with fields for username and password.
	2	Driver enters their assigned username and password.
	3	Driver clicks the 'Login' button.
	4	System validates the credentials against the user database.
	5	System confirms credentials are valid and grants access
	6	System redirects the driver to their role-specific dashboard.
Extensions	Step	Branching Action
	4a	Condition: Username is not found. Action: System displays an "Invalid username or password" error.
	4b	Condition: Password is incorrect. Action: System displays an "Invalid username or password" error.

Use Case DRI-02: Starting the trip

Number	DRI-02
Name	Starting the trip
Summary	The driver starts a trip log to record journey details.
Priority	5
Preconditions	1. Driver must be logged into the system. 2. Assigned to a bus.
Postconditions	1. Trip log is created. 2. Journey details are recorded.
Primary Actor(s)	Bus Driver
Secondary Actor	System Admin

Trigger	Driver selects 'Start Trip Log' option	
Main Scenario	Step	Action
	1	Driver logs into the system.
	2	Driver navigates to trip management.
	3	Driver selects 'Start Trip Log'.
	4	System creates a new trip log entry.
Extensions	Step	Branching Action
	3a	Condition: If the driver is not assigned to a route Action: System shows an error message.

Use Case DRI-03: View Route Details

Number	DRI-03
Name	View Route Details
Summary	The driver views their assigned route details to follow the correct schedule.
Priority	5
Preconditions	1. Driver must be logged in 2. Driver must be assigned to a route.
Postconditions	Driver views and confirms assigned route details.
Primary Actor(s)	Bus Driver
Secondary Actor	System Admin

Trigger	Driver selects 'View Route Details' option.	
Main Scenario	Step	Action
	1	Driver logs into the system.
	2	Driver navigates to route management.
	3	Driver selects 'View Route Details'.
	4	System displays the assigned route information.
Extensions	Step	Branching Action
	3a	If no route is assigned, system notifies driver and prompts to contact operations staff.

Use Case DRI-04: Update Route Details

Number	DRI-04
Name	Update Route Details
Summary	The driver updates the trip status to reflect journey progress.
Priority	4
Preconditions	3. Driver must be logged in 4. Driver must be assigned to a route.
Postconditions	Driver views and confirms assigned route details.
Primary Actor(s)	Bus Driver
Secondary Actor	System Admin

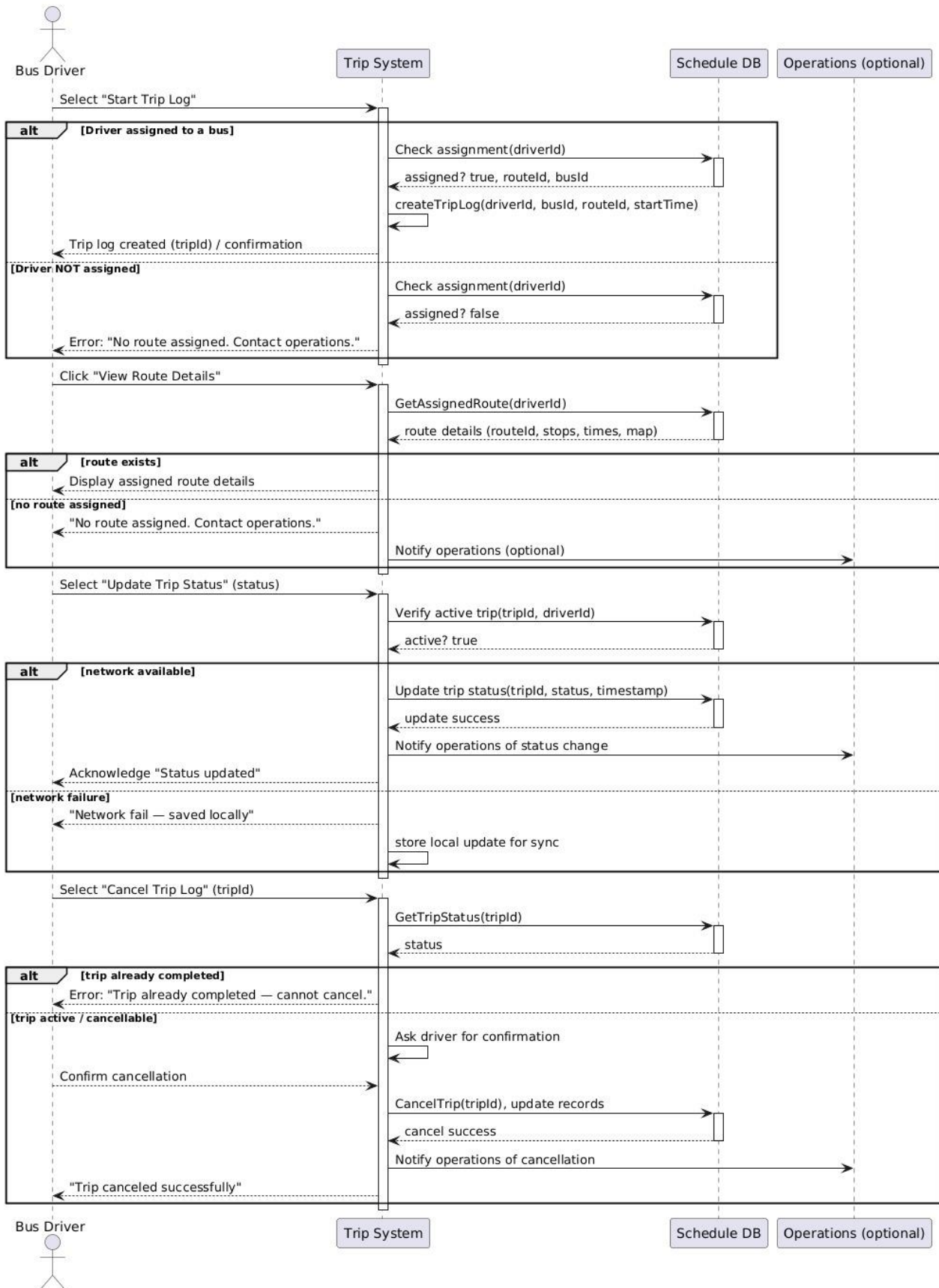
Trigger	Driver selects 'Update Trip Status' option.	
Main Scenario	Step	Action
	1	Driver logs into the system.
	2	Driver navigates to route management.
	3	Driver selects 'UpdateRoute Details'.
	4	Driver updates the status (e.g., Started, In Progress, Completed).
	5	System updates and notifies operations staff.
Extensions	Step	Branching Action
	4a	Condition: If network connection fails Action: update is stored locally and synced when available

Use Case DRI-05: Cancel Trip Log

Number	DRI-05
Name	Cancel Trip Log
Summary	The driver cancels a trip log when a trip is not completed.
Priority	4
Preconditions	Driver must have an active trip log.
Postconditions	Trip log is canceled and records are updated.
Primary Actor(s)	Bus Driver
Secondary Actor	System Admin

Trigger	Driver selects 'Cancel Trip Log' option.	
Main Scenario	Step	Action
	1	Driver logs into the system.
	2	Driver navigates to trip log management.
	3	Driver selects 'Cancel Trip Log'.
	4	System cancels the trip log and updates records.
Extensions	Step	Branching Action
	3a	If trip log has already been completed, cancellation is not allowed.

DRI-05 Cancel Trip Log



IT24102501-Bandara C.V

Use Case Scenarios for Actor: Company Owner Use

Case CO-01: Login to System

Number	CO-01
Name	Login to System
Summary	The Company Owner provides credentials to gain authenticated access to the system dashboard.
Priority	5
Preconditions	1. The company owner has a valid user account in the system. 2. The company owner is not already logged in.
Postconditions	1. The company owner is authenticated. 2. The system dashboard for the company owner role is displayed.
Primary Actor(s)	Company Owner
Secondary Actor	-

Trigger	The company owner navigates to the system login page.	
Main Scenario	Step	Action
	1	System displays the login form with fields for username and password.
	2	Company owner enters their assigned username and password.
	3	Company owner clicks the 'Login' button.
	4	System validates the credentials against the user database.
	5	System confirms credentials are valid and grants access
	6	System redirects the company owner to their role-specific dashboard.
Extensions	Step	Branching Action
	4a	Condition: Username is not found. Action: System displays an "Invalid username or password" error.

	4b	Condition: Password is incorrect. Action: System displays an "Invalid username or password".
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Use Case CO-02: Add new routes

Number	CO-02
Name	Add new routes
Summary	The company owner adds new routes to the system.
Priority	5
Preconditions	1. The company owner is logged in. 2. The company owner has access rights to manage buses.
Postconditions	1. The new route is registered in the system.
Primary Actor(s)	Company Owner
Secondary Actor	-

Trigger	The bus owner selects 'Add New Route.	
Main Scenario	Step	Action
	1	Owner navigates to the bus management dashboard.
	2	Owner selects 'Add New Route.
	3	System displays the add the route.
	4	Owner enters new route details.
	5	Owner submits the form.
	6	System validates inputs and registers the new route.
	7	System confirms the route has been added.
Extensions	Step	Branching Action
	4a	Condition: Invalid or missing details. Action: System displays an error and requests correction.

Use Case CO-03: View Route Details

Number	CO-03
Name	View Route Details
Summary	The company owner views all buses to monitor their operational fleet.
Priority	5
Preconditions	1. The ompany owner is logged in.
Postconditions	1. . The system displays details of all routes.
Primary Actor(s)	Company Owner
Secondary Actor	-

Trigger	The company owner selects 'View Route Details'.	
Main Scenario	Step	Action
	1	Owner navigates to the dashboard.
	2	Owner selects 'View Route Details'.
	3	System retrieves all route records.
	4	System displays bus information.
	5	Owner reviews the route details.
Extensions	Step	Branching Action
	3a	Condition: No buses are available. Action: System displays an empty list message.

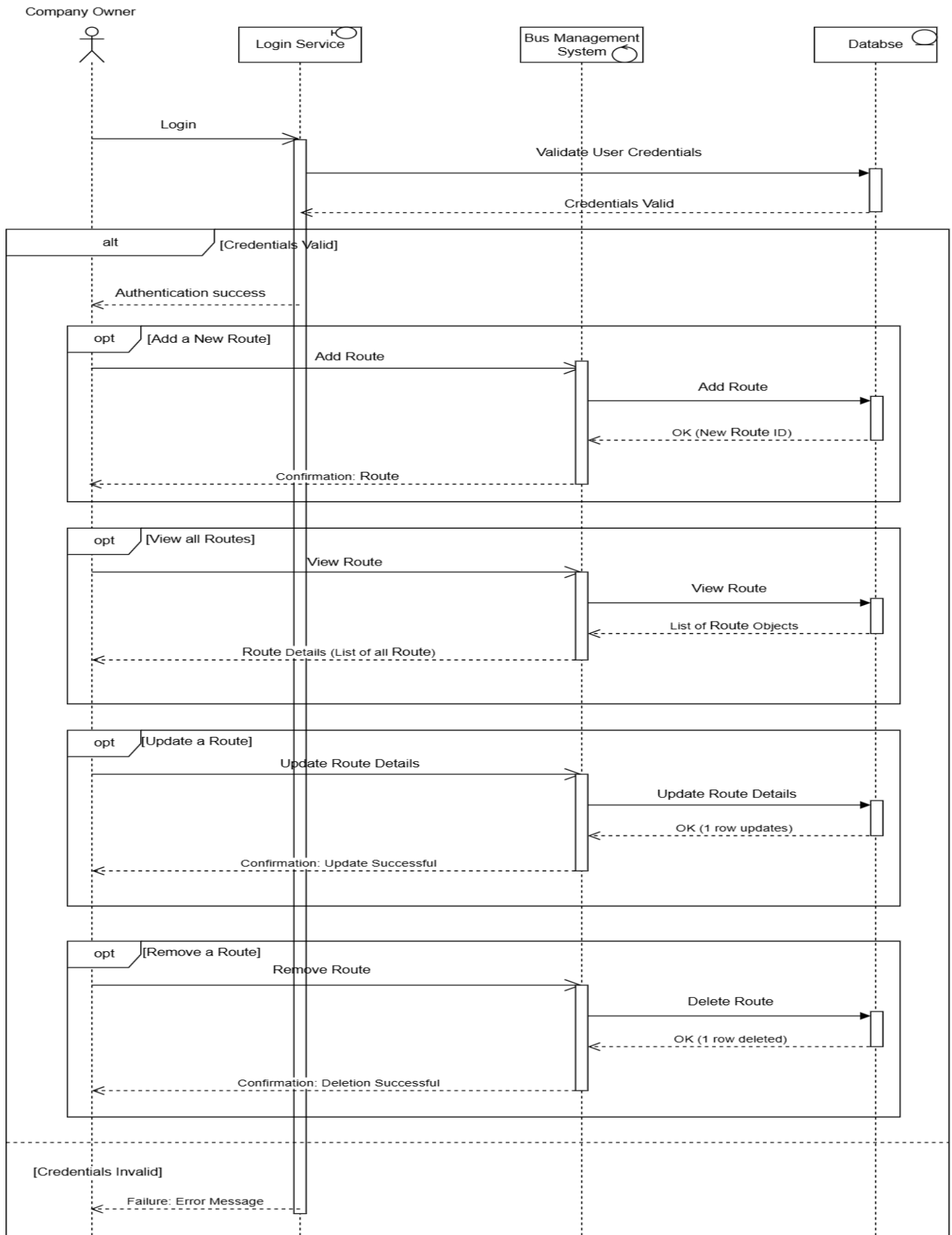
CO-04: Update Route Details

Number	CO-04	
Name	Update Route Details	
Summary	The company owner updates bus details such as seating capacity or license information.	
Priority	4	
Preconditions	1. The company owner is logged in. 2. The company exists in the system.	
Postconditions	1. The route record is updated with new detail	
Primary Actor(s)	Company Owner	
Secondary Actor	-	
Trigger	The bus owner selects 'Update route Details'.	
Main Scenario	Step	Action
	1	Owner navigates to the bus management dashboard.
	2	Owner selects the route to update.
	3	System displays current details of the route.
	4	Owner edits the needed information.
	5	Owner submits the updates.
	6	System validates and saves the changes.
	7	System confirms the route has been updated.
Extensions	Step	Branching Action
	4a	Condition: Invalid or missing details. Action: System displays an error and requests correction.

CO-06: Remove Route

Number	CO-06
Name	Remove route
Summary	The ompany owner removes a route from the system so that only active buses remain listed.
Priority	4
Preconditions	1. The ompany owner is logged in. 2. The route exists in the system.
Postconditions	1. The route record is removed from the system.
Primary Actor(s)	Bus Owner
Secondary Actor	-

Trigger	The company owner selects 'Remove route route.	
Main Scenario	Step	Action
	1	Owner navigates to the route management dashboard.
	2	Owner selects a route to remove.
	3	System displays the route details and a remove option.
	4	Owner confirms removal.
	5	System deletes the route record.
	6	System confirms the route has been removed.
Extensions	Step	Branching Action
	2a	The selected route is currently active in a journey. Action: System prevents deletion and shows a warning.



IT24102866-Kalendra G.K.A

Use Case Scenarios for Actor: System Administrator Use

Case SYS-01: Login to System

Number	SYS-01
Name	Login to System
Summary	The system administrator provides credentials to gain authenticated access to the admin dashboard.
Priority	5
Preconditions	1. The admin has a valid user account. 2. The admin is not already logged in.
Postconditions	1. The admin is authenticated. 2. The admin dashboard is displayed.
Primary Actor(s)	System Administrator
Secondary Actor	-

Trigger	The admin navigates to the system login page.	
Main Scenario	Step	Action
	1	System displays the login form.
	2	Admin enters their username and password.
	3	Admin clicks the 'Login' button.
	4	System validates the credentials against the admin database.
	5	System confirms credentials are valid and grants access.
	6	System redirects the admin to the administrator dashboard.
Extensions	Step	Branching Action
	4a	Condition: Invalid credentials. Action: System displays an "Invalid username or password" error.

Use Case SYS-02: Manage User Accounts

Number	SYS-02
Name	Manage User Accounts
Summary	The admin creates, views, updates, or deactivates user accounts for all roles (Driver, Conductor, Bus Owner, etc.).
Priority	5
Preconditions	Admin is logged into the system
Postconditions	The user account is created, updated, or deactivated as intended.
Primary Actor(s)	System Administrator
Secondary Actor	-

Trigger	Admin navigates to the 'User Management' section from the dashboard.	
Main Scenario	Step	Action
	1	Admin selects 'User Management' from the dashboard.
	2	System displays a list of all user accounts.
	3	Admin chooses to Create, View, Edit, or Deactivate an account.
	4	Admin provides/edits the necessary user information (name, role, contact details, login credentials).
	5	Admin saves the changes.
	6	System validates and updates the user database.
	7	System confirms the action was successful.
Extensions	Step	Branching Action
	4a	Condition: Admin tries to create a user with an existing username/email. Action: System displays a "User already exists" error.
	6a	Condition: Required fields are missing. Action: System prompts the admin to fill in all required fields.

Use Case SYS-03: Manage Driver/Conductor Schedule

Number	SYS-03
Name	Manage Driver/Conductor Schedule
Summary	The admin assigns drivers and conductors to specific buses and routes for a given date and time.
Priority	4
Preconditions	1. Admin is logged in. 2. The bus, driver, conductor, and route exist in the system.
Postconditions	A new schedule is created or an existing one is modified
Primary Actor(s)	System Administrator
Secondary Actor	-

Trigger	Admin navigates to the 'Schedule Management' section.	
Main Scenario	Step	Action
	1	Admin selects 'Schedule Management'.
	2	System displays a list of existing schedules and an option to create a new one.
	3	Admin selects a driver, conductor, bus, route, date, and time.
	4	System checks for scheduling conflicts.
	5	Admin saves the new schedule.
	6	System confirms the schedule has been saved.
Extensions	Step	Branching Action
	4a	Condition: A conflict is found (e.g., driver already assigned). Action: System displays a warning and prevents saving.
	5a	Condition: Admin attempts to assign a deactivated user. Action: System displays an error: "Cannot assign a deactivated user."

Use Case SYS-04: View System Logs

Number	SYS-04
Name	View System Logs
Summary	The admin views system logs for monitoring, auditing, and troubleshooting
Priority	3
Preconditions	Admin is logged in.
Postconditions	The requested log data is displayed.
Primary Actor(s)	System Administrator
Secondary Actor	-

Trigger	Admin navigates to the 'System Logs' section.	
Main Scenario	Step	Action
	1	Admin selects 'System Logs'.
	2	System displays filter options (e.g., date, event type, user).
	3	System retrieves and displays the log entries.
	4	Admin can review the logs.
Extensions	Step	Branching Action
	4a	Condition: No log entries match the filter criteria. Action: System displays a "No logs found" message.

Use Case SYS-05: Generate Technical Reports

Number	SYS-05
Name	Generate Technical Reports
Summary	The admin generates reports on system usage, errors, and performance.
Priority	3
Preconditions	Admin is logged in.
Postconditions	The report is generated and displayed.
Primary Actor(s)	System Administrator
Secondary Actor	-

Trigger	Admin navigates to the 'Reports' section and selects 'Technical Reports'	
Main Scenario	Step	Action
	1	Admin selects report type and date range.
	2	Admin clicks 'Generate Report'.
	3	System queries the database and log files.
	4	System compiles and displays the report (e.g., number of errors, user login frequency).
	5	System offers an option to export the report.
Extensions	Step	Branching Action
	3a	Condition: No data is available for the selected criteria. Action: System displays a "No data available" message.

